

# SUMIT SAH

San Marcos, TX | Email | LinkedIn | GitHub | Portfolio

## Education

### Texas State University, Honors College

Expected May 2027

B.S. Double Major: Applied Mathematics & Computer Science — GPA: 3.85

San Marcos, TX

- **Relevant Coursework:** Machine Learning, Probability & Statistics, Data Structures & Algorithms, Technical Writing
- **Honors & Awards:** Outstanding Undergraduate Student Award; President's Honor Scholarship; President's List (4x)

## Experience

### Machine Learning Engineering Intern

May 2026 – Aug 2026

Microsoft Corporation, ODSP-DEEDS

Redmond, WA

- Developed and evaluated vectorless RAG pipelines (retrieval without dense embeddings), including LLM-Wiki and PageIndex, in offline experiments benchmarking retrieval approaches for enterprise-scale content grounding in Microsoft Copilot.
- Conducted recall-focused experiments across query sets and retrieval configurations, increasing Recall@K from a 68% baseline to as high as 96% with the best-performing approach.
- Built an LLM-as-a-judge evaluation pipeline using assertion-based checklists to assess generated responses, achieving 100% completeness and up to 89% response quality on targeted evaluation sets.
- Explored LLM-based query generation, reformulation, and container summarization to improve semantic search coverage across complex document hierarchies.

### Undergraduate Research Assistant

Apr 2025 – Present

Department of Mathematics & Computer Science, Texas State University

San Marcos, TX

- Conducted research on optimal transport-based distributionally robust optimization (OT-DRO) for robust regression, benchmarking against OLS, ridge, and lasso under noise, outliers, distribution shift, and sparsity settings using Python and PyTorch.
- Presented research findings through conference poster presentations and invited departmental talks, communicating theoretical insights and experimental results on robust regression and optimal transport to academic audiences.

### Data Analysis & Visualization Assistant

Sep 2025 – Present

University Libraries – Research & Data Services, Texas State University

San Marcos, TX

- Delivered 1:1 consultations and workshops on Python, R, and statistical methods, helping students and faculty apply data analysis techniques to research and coursework.
- Co-organized the TXST Datathon, Data & Donuts, and Open House programming; supported marketing and outreach efforts that increased student participation and engagement.
- Maintained Research & Data Services LibGuides and library web resources, updating back-end content and improving access to data tools, tutorials, and services.

## Projects

### AI-Powered Study Assistant | React, Flask, Firebase, OpenAI API, Qdrant | GitHub

- Built an AI-powered study platform that generates quizzes from course notes, identifies weak topics, and tracks learning progress through personalized study workflows.
- Designed a semantic retrieval pipeline with Qdrant vector search to surface relevant course-note context before OpenAI-powered quiz generation.
- Implemented real-time Firestore synchronization and unit tests for quiz tracking, history retrieval, answer review, and retake workflows.

### RateMyProfessor Bias Analysis – TXST Datathon 2025, 3rd Place | GitHub

- Analyzed 20K+ RateMyProfessor reviews with Python and Pandas to identify grading, sampling, and representation biases in student evaluations.
- Created 5+ visualizations and proposed actionable approaches to improve rating reliability; earned 3rd place among 10 teams.

## Skills

- **AI/ML:** PyTorch, Scikit-learn, RAGs, LLM-as-a-judge, LangChain, LLM APIs, vector DB, Qdrant, Deep Learning, statistical modeling, Hugging Face (Transformers, Datasets, PEFT)
- **Programming & Data:** Python, R, SQL, C/C++, Java, Pandas, NumPy, Matplotlib, Tableau, Power BI, MATLAB
- **Optimization:** CVXPY, scipy.optimize
- **Tools & Deployment:** Git, Linux, Docker, Flask, Firebase, Streamlit, CI/CD

## Leadership & Extracurriculars

- **Math Club @ TXST – President:** Lead a 100+ member organization and organize technical and academic events, including AI talks, workshops, Calculus Bee, Escape Room, and Pi Day.
- **Content Creation:** Produced 30+ educational YouTube videos using Manim (1.1K+ subscribers, 50K+ views) and grew an Instagram education channel to 6K+ followers and 1M+ views across math, CS, and AI content.
- **SIAM & ACM AI:** Participated in 6+ technical workshops, Kaggle competitions, and 3 hackathons focused on AI ethics and neural networks.